MQ135 Gas Sensor

1. Sensor Overview and Purpose

The MQ135 is an analog chemoresistive gas sensor broadly sensitive to a range of polluting gases, including Ammonia (NH3), Nitrogen Oxides (NOx), Alcohol, Benzene, Smoke, and Carbon Dioxide (CO2). In this system, the MQ135 is used to measure General Gas Pollution Intensity on a normalized 0–100% scale.

The MQ135 requires a 5V power supply to heat its internal sensing layer. So, its analog output pin (A0) can output voltages ranging from 0.1V up to nearly 5.0V.

Because the ESP32-S3 microcontroller's Analog-to-Digital Converter (ADC) pins are strictly limited to a maximum of 3.3V, directly connecting the sensor would risk hardware. To safely interface the sensor, a Voltage Divider circuit was implemented on the PCB. Using two resistors of equal value (R1 and R2 at 10kΩ each), the 5V signal is safely stepped down to a maximum of 2.5V before reaching the ESP32 GPIO pin. The ESP32 code then mathematically multiplies the measured voltage by 2 to reconstruct the true sensor output voltage.

2. Calibration Procedure (R0 Calculation)

According to the MQ135 datasheet, the resistance ratio (Rs/R0) in clean, fresh air is a constant 3.6.

- Step 1: Turn your device on and let it run for at least 30 minutes, because the sensor requires a warm-up period. When first powered on, the readings will fluctuate significantly until its internal temperature stabilizes.
- Step 2: Take your powered-on device outside into fresh air. Let it sit there for about 5 minutes to adjust to the outdoor air.
- Step 3: Over the next 5 to 10 minutes, record the Rs (Sensor Resistance) values from an Arduino code and take the average value of them.
- Step 4: Take that average Rs and simply divide it by 3.6 (the clean-air rule from the manufacturer's datasheet). We got the R0 as 33.62567 kΩ.

$$\frac{R_s}{R_0} = 3.6$$

3. Calculate the "Gas Pollution Percentage"

The Fixed Parameters

- R0: The calibrated baseline resistance in clean air.
- R(load): This is the physical 10kΩ resistor soldered onto the back of the MQ135 module.
- a(PARA): 102.2 (A scaling number taken directly from the Ammonia/Smoke sensitivity curve in the MQ135 datasheet).

- b(PARB): 2.473 (A slope number taken directly from that exact same datasheet curve).

The equations used in the code

Equation 1: The ESP32's analog pin reads the sensor as a digital number between 0 and 4095. The code must convert this back into a physical voltage (out of the 3.3V maximum).

$$V_{pin} = Raw \times \left(\frac{3.3}{4095}\right)$$

Equation 2: Because your PCB has two resistors dividing the voltage in half to protect the ESP32, the sensor's actual output is twice as high as what the ESP32 sees.

$$V_{true} = V_{pin} \times 2.0$$

Equation 3: Finding the Current Sensor Resistance (Rs)

Now the code calculates the real-time electrical resistance of the chemical sensor (Rs). Since the MQ135 is powered by 5.0V, it uses the standard voltage divider formula alongside your R(load) parameter:

$$R_s = R_{load} \times \left(\frac{5.0 - V_{true}}{V_{true}}\right)$$

Equation 4: This is the core datasheet equation. The code takes the current air (Rs), divides it by your clean air memory (R0), and applies the datasheet curve parameters (a and b) to find the exact concentration of gas in Parts Per Million.

$$PPM = a \times \left(\frac{R_s}{R_0}\right)^{-b}$$

Equation 5: Finally, the code converts that PPM number into a 0% to 100% scale using a linear mapping formula (which Arduino handles using the map() function). If we define 0 PPM as 0% and 10 PPM as 100%:

$$Percentage = \left(\frac{PPM - 0}{10 - 0}\right) \times 100$$

Here is what the different output percentages actually mean:

- 0% to 10% (Fresh Air): The air is perfectly clean and fresh. The code calculates around 0 to 1 PPM.
- 20% to 40% (Mild Pollution): Trace amounts of general polluting gases have entered the air. The code calculates around 2 to 4 PPM.
- 50% to 70% (Poor Air Quality): There is a strong, distinct presence of smoke or chemical gas in the air. The code calculates around 5 to 7 PPM.
- 80% to 100% (Toxic Level): The air is heavily polluted. The code calculates around 8 to 10 PPM.